

Substitution reactions involving radicals and alkanes

Learning outcomes

By the end of this section you should be able to explain, using equations, the initiation, propagation and termination steps in the reactions between alkanes and halogens.

Earlier in your studies you learned about plastic polymers such as polyethylene (both high density and low density) and polypropylene (see [section S2.4.4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/>)). These polymers are examples of [alkanes](#) (see [section S3.2.3–4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>)) which are organic compounds made up of only single-bonded carbon and hydrogen atoms with no other functional groups. These alkane polymers are shown in **Figure 1** along with some of their uses.

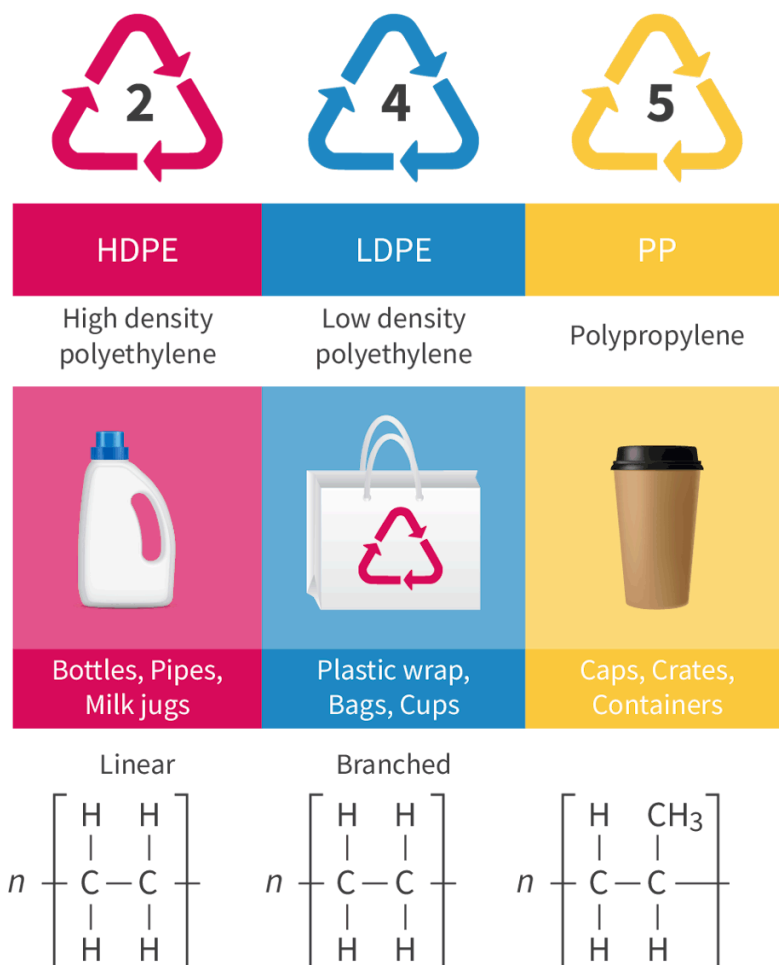


Figure 1. Alkane plastic polymers.

 More information for figure 1

The image is a diagram displaying three types of plastic polymers: HDPE, LDPE, and PP. Each type is shown with a recycling symbol and code number: HDPE (2), LDPE (4), and PP (5).

HDPE (High-density polyethylene) has a linear structure and is shown with an image of a bottle, representing products like bottles, pipes, and milk jugs.

LDPE (Low-density polyethylene) has a branched structure and is associated with plastic wrap, bags, and cups, illustrated with a bag icon.

PP (Polypropylene) is depicted with a cup, signifying items such as caps, crates, and containers, and it displays a structure with a CH₃ group.

The diagram includes molecular structures: the HDPE and LDPE show repeating units of H attached to a carbon backbone, while PP shows methyl groups (CH₃) attached.

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We know plastic polymers are a global concern, one of the reasons being that they do not break down easily and remain intact for many, many years. What does that suggest about the reactivity of these plastic alkane polymers and what do you think that means about the reactivity of alkanes?

Can you think of any species that might be able to react with these very unreactive alkanes?

In this section you will learn about a specific type of reaction involving alkanes known as a substitution reaction. When you think of the word 'substitution', what comes to mind? Consider making a substitution in a meal you order from a restaurant or making a substitution in a game of volleyball. What does the word substitution mean in these contexts? What do you think a substitution reaction is?

Alkanes

Why do you think alkanes are very unreactive and stable species? What is it about the structure of alkanes that make them so unreactive?

There are two main reasons as to why alkanes are relatively unreactive species both centred around the covalent bonds that make up alkanes.

- **Non-polar bonds:** The carbon–carbon (C–C) and carbon–hydrogen (C–H) bonds are non-polar in nature, meaning there is a relatively symmetrical distribution of electrons within the

covalent bond. This means there are no partial charges in the molecule, limiting any attraction from other molecules.

- **Relatively strong bonds:** The C–C and C–H bonds have a high bond enthalpy and require a lot of energy in order to break.

Table 1 shows a comparison of some bond enthalpies.

Table 1. Bond enthalpies for some carbon containing covalent bonds.

Bond	Bond enthalpy (kJ mol ^{−1})
C–H	414
C–C	346
C–Br	285
C–N	286

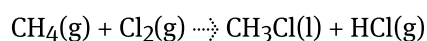
Making connections

Alkanes tend to be unreactive, indicating that there is a high activation energy to overcome for most reactions. However, alkanes do undergo combustion reactions in the presence of oxygen. Since combustion reactions are exothermic this would mean the products during combustion are more stable than the reactants.

Thinking back to what you learned in [sections R1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/>) and [R2.2.4–5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>), why do you think alkanes are described as kinetically stable but thermodynamically unstable?

Radical substitution reactions

Can you think of any species that are highly reactive that might be able to react with alkanes? Radicals! Alkanes can undergo a type of reaction called a substitution reaction, where a part of the alkane is substituted by a radical. An example of a radical substitution reaction would involve an alkane and a halogen reacting to form a halogenoalkane. The reaction of methane and chlorine is such a reaction:



Radical substitution reactions always involve three steps (**Table 2**):

- **Initiation** – formation of a radical.
- **Propagation** – radical species reacts with a covalent species forming a new radical and a new covalent species. Since another radical is formed this step causes a chain reaction.
- **Termination** – two radicals form a single covalent species, terminating the chain reaction.

Table 2. Reactions involved in the initiation, propagation and termination steps of the radical substitution of methane with chlorine.

Step	Reaction(s)	Notes
Initiation	$\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$	Endothermic (UV radiation or heat required).
Propagation	$\text{CH}_4 + \text{Cl}\cdot \rightarrow \cdot\text{CH}_3 + \text{HCl}$	Chain reaction (exothermic or endothermic depending on bond energies).
	$\cdot\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$	
	$\text{CH}_3\text{Cl} + \text{Cl}\cdot \rightarrow \cdot\text{CH}_2\text{Cl} + \text{HCl}$	
	$\cdot\text{CH}_2\text{Cl} + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{Cl}\cdot$	
Termination	$\cdot\text{CH}_3 + \text{Cl}\cdot \rightarrow \text{CH}_3\text{Cl}$	Exothermic (formation of a stable bond, unstable reactive species).
	$\cdot\text{CH}_3 + \cdot\text{CH}_3 \rightarrow \text{C}_2\text{H}_6$	
	$\text{Cl}\cdot + \text{Cl}\cdot \rightarrow \text{Cl}_2$	

What do you notice about the products formed from these reactions? Is there one overall product or are there many possible options of products that can form? If you notice, the propagation step involves a radical as a reactant, but it also produces another radical as a product. Since radicals are highly reactive species both the radical produced and the initial radical are able to react. This means that during a radical substitution reaction, there is a mixture of products formed.

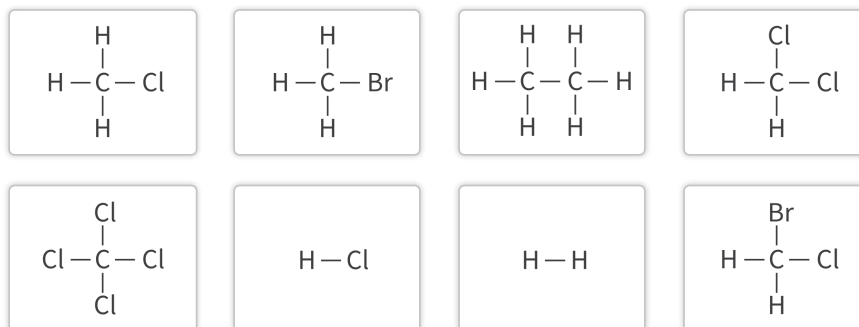
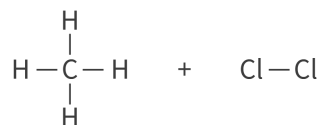
We can look more closely at the reaction mechanism for these three steps in the radical substitution reactions of alkanes. **Table 3** shows the movement of the single electron using a single-barbed arrow in the reaction mechanism.

Table 3. Reaction mechanism for the initiation, propagation and termination steps of the radical substitution of methane with chlorine.

Step	Reaction(s)	Notes
Initiation	$\text{Cl}-\text{Cl} \xrightarrow{\text{UV}} \text{Cl}\cdot + \cdot\text{Cl}$ ☉ More information	Endothermic radiation or heat required).
Propagation	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array} + \cdot\text{Cl} \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\cdot \\ \\ \text{H} \end{array} + \text{H}-\text{Cl}$ ☉ More information	Chain reaction or endothermic depending on energies).
	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\cdot \\ \\ \text{H} \end{array} + \text{Cl}-\text{Cl} \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{Cl} \\ \\ \text{H} \end{array} + \cdot\text{Cl}$ ☉ More information	
Termination	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\cdot \\ \\ \text{H} \end{array} + \cdot\text{Cl} \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{Cl} \\ \\ \text{H} \end{array}$ ☉ More information	Exothermic (formation of a stable bond from two unstable reactive species).
	$\text{Cl}\cdot + \cdot\text{Cl} \longrightarrow \text{Cl}-\text{Cl}$ ☉ More information	

You may also notice in the initiation and propagation steps the formation of alkyl and halogen radicals. The formation of a hydrogen radical, while not impossible, is rare. The combination of hydrogen with a halogen radical to form a stable covalent bond is a much more probable process than the formation of a hydrogen radical.

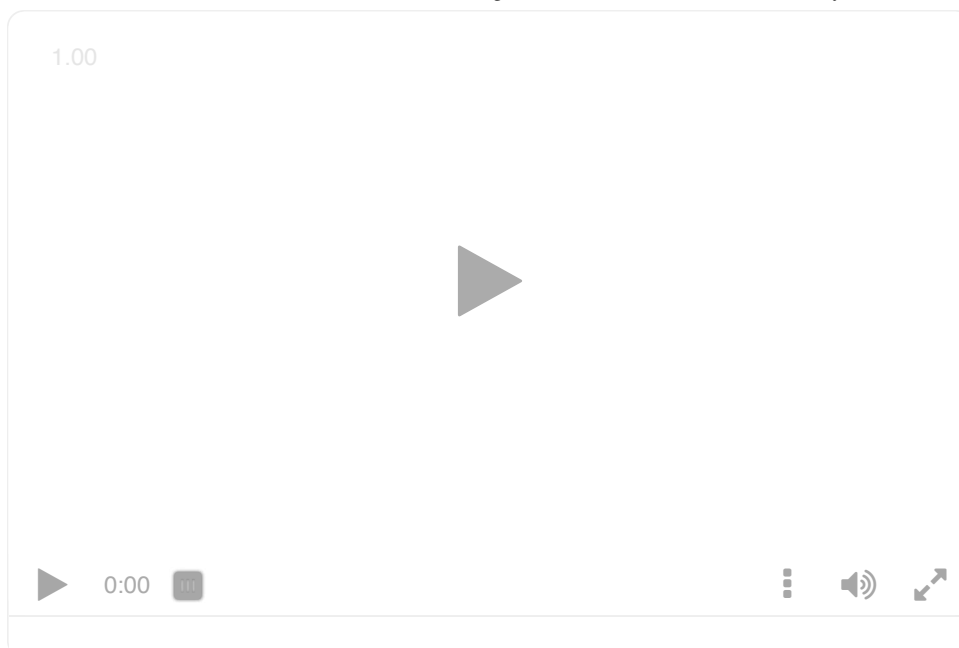
Use **Interactive 1** to predict what products are possible for the radical substitution reaction involving methane and chlorine.



✓ Check

Interactive 1. What products are possible for the radical substitution reaction involving methane and chlorine?

In section R3.3.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/radicals-and-how-they-are-produced-id-43489/>) we looked at the reaction mechanism involving single-barbed arrows for the formation of a halogen free radical. The reaction mechanism for each step of the radical substitution of alkanes will also involve a single-barbed arrow. **Interactive 2** shows the reaction mechanism for the first reaction of each step shown in **Table 2**.



Interactive 2. Reaction Mechanism for the Free Radical Substitution of Methane.

 More information for interactive 2

A video displays the reaction Mechanism for the Free Radical Substitution of Methane.

Step: Initiation. Important notes: Requires UV radiation or heat. It starts with a Cl_2 molecule with two Cl atoms, each with a valence electron. An arrow points from each of the electrons toward the Cl atom. A chemical reaction displays the Cl_2 molecule leading to two Cl atoms with valence electrons.

Step: Propagation. Important notes: Chain reaction. Reaction shows C bonded to four H atoms plus Cl atom with an electron. One of the bonds is depicted as a pair of electrons with two dots between C and H. An arrow points from one electron toward a C. There is an arrow pointing from the valence electrons from the H and Cl atoms toward each other. It leads to C having an electron and is single-bonded to three H atoms. H single bond Cl.

Step: Termination. Important notes: Mixture of products end up forming. A chemical reaction. C with an electron single bonded to three H atoms. Cl with an electron. Arrows point from the electrons of C and Cl towards each other. It yields C single bonds to three H atoms and a Cl atom.

Radical substitution of an alkanes with a halogen is a common starting place to produce the more reactive halogenoalkane from the less reactive alkane. This is a helpful first step for chemists in the synthesis of molecules with specific functional groups, starting with an alkane species.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly

- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Group activity (3—5 students)

In groups you will complete the downloadable worksheet while sitting in a circle with your group. You will rotate your sheet clockwise after each round. After the final round you will collect your sheet back. There should be no talking to group members through rounds 3—9 to put an emphasis on using terminology, symbols and conventions to communicate your ideas on the page.

Worksheet 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/ChemistryWorksheet.pdf)

Round 1 — Select an alkane of your choosing. Ensure each member of your group selects a different alkane. Draw the structure of your chosen alkane in the space provided. Pass your sheet clockwise and gather a new sheet from the person to your right.

Round 2 — Select a halogen of your choosing. Ensure each member of your group selects a different halogen. List your halogen in the space provided. Rotate sheets clockwise.

Round 3 — Write the chemical equation for the initiation step and state any conditions required for this step in the space provided. Rotate sheets clockwise.

Round 4 — Write a chemical equation for a propagation step in the space provided. Rotate sheets clockwise.

Round 5 — Write another chemical equation for a propagation step in the space provided. Rotate sheets clockwise.

Round 6 — Write one more chemical equation for a propagation step in the space provided. Rotate sheets clockwise.

Round 7 — Write a chemical equation for a termination step in the space provided. Rotate sheets clockwise.

Round 8 — Write another chemical equation for a termination step in the space provided. Rotate sheets clockwise.

Round 9 — Write one more chemical equation for a termination step in the space provided. Rotate sheets clockwise.

After all nine rounds have been completed, return each sheet to its original owner. Individually, go through the work from each round and correct any errors you to identify with a different coloured pen. Discuss as a group which round you found the most difficult and which round you found the most straightforward and why.